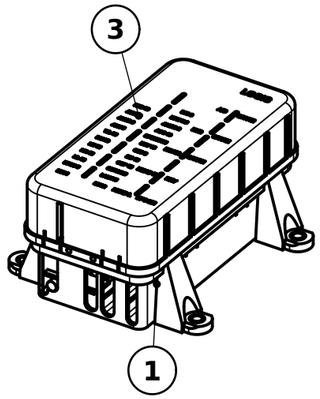
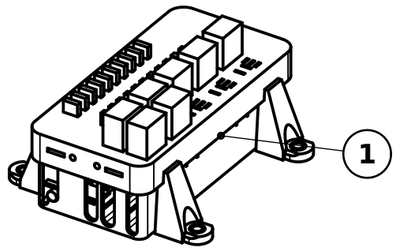
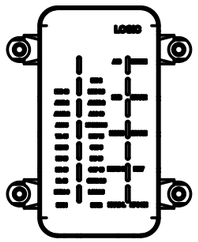
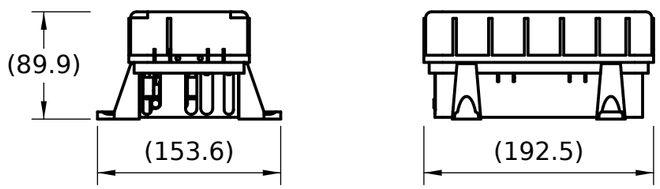


TRIMETRIC VIEW



TRIMETRIC - COVER REMOVED





FRONT / TOP / RIGHT VIEWS SCALE 1:5

REVISIONS

REV	DESCRIPTION	BY	DATE
P	#273: FB-105 base M, FB-106 pan F, FB-107 cover H, FB-115 / -116 strips D	Claude	16/10/26
R	#275: FB-105 base N	Claude	17/10/26
S	#297: FB-105 base P, FB-106 pan G, FB-107 cover J, FB-115 / -116 strips E	Claude	18/10/26
T	#303: FB-105 base R, FB-106 pan H, FB-107 cover K	Claude	18/10/26
U	#308: FB-105 base S; first assembly drawing (buildability review 18/10/26)	Claude	18/10/26
V	FB-105 base T (strip hold-downs)	Claude	18/10/26

ITEM	QTY	PART NUMBER	DESCRIPTION	MATERIAL / SPEC
1	1	ST185-FB-105	LOGIC BASE	PA12, 3D PRINTED
2	1	ST185-FB-106	LOGIC PAN (BELLY PAN)	PA12, 3D PRINTED
3	1	ST185-FB-107	LOGIC COVER	PA12, 3D PRINTED
4	1	ST185-FB-115/-116	LOGIC BUS STRIPS BUS1 + BUS2	C110 Cu, TINNED
5	19	PEM S-M4-2ZI	SELF-CLINCHING NUT M4, PRESSED INTO THE STRIP	C110 Cu, TINNED
6	2	PEM SS-M5-2ZI	SELF-CLINCHING NUT M5, PRESSED INTO THE STRIP	C110 Cu, TINNED
7	4	ATM-2	MINI BLADE FUSE 2 A (ATM)	PURCHASED
8	3	ATM-5	MINI BLADE FUSE 5 A (ATM)	PURCHASED
9	4	ATM-7-1/2	MINI BLADE FUSE 7.5 A (ATM)	PURCHASED
10	2	ATM-10	MINI BLADE FUSE 10 A (ATM)	PURCHASED
11	7	ATM-15	MINI BLADE FUSE 15 A (ATM)	PURCHASED
12	1	ATM-20	MINI BLADE FUSE 20 A (ATM)	PURCHASED
13	1	4-1904124-2	MICRO RELAY (ALL 7 POSITIONS)	PURCHASED
14	1	4-1904124-3	MICRO RELAY: HORN (SEE NOTE 2) (NOT SHOWN)	PURCHASED
15	2	5-1393292-8	MICRO RELAY: HEAD LH, HEAD RH (SEE NOTE 2) (NOT SHOWN)	PURCHASED
16	3	6-1419137-4	MICRO RELAY: K_ACC, K_EPS_EN, K_REV (SEE NOTE 2) (NOT SHOWN)	PURCHASED
17	14	280919-2	MICRO RELAY COIL RECEPTACLE 4.8 X 0.8	PURCHASED
18	17	42281-1	MICRO RELAY LOAD RECEPTACLE 6.3 X 0.8	PURCHASED
19	22	1-968849-1	MINI FUSE RECEPTACLE MCP 2.8, 20 / 18 AWG	PURCHASED
20	20	1-968851-1	MINI FUSE RECEPTACLE MCP 2.8, 16 / 14 AWG	PURCHASED
21	1	RNF-100-1/8-0-STK	INSULATION BUILD-UP SLEEVE 1/8 (NOT SHOWN)	PURCHASED
22	1	RNF-100-3/32-0-STK	INSULATION BUILD-UP SLEEVE 3/32 (NOT SHOWN)	PURCHASED
23	4	94459A140	M3 HEAT-SET INSERT (OPTIONAL COVER BACKUP) (NOT SHOWN)	BRASS
24	4	SSCF-M3-10-A4	M3 x 10 CAP SCREW (OPTIONAL COVER BACKUP) (NOT SHOWN)	A4 STAINLESS
25	11	320554	INPUT JUMPER RING LUG, 22-16 AWG, #8 / M4	Cu, TINNED
26	8	53941-1	INPUT JUMPER RING LUG, 16-14 AWG, #8 / M (SEE NOTE 2) (NOT SHOWN)	Cu, TINNED
27	1	M22759/16-14-2	INPUT JUMPER WIRE 14 AWG, ~60 MM EACH	PURCHASED
28	1	M22759/16-16-2	INPUT JUMPER WIRE 16 AWG, ~60 MM EACH (SEE NOTE 2) (NOT SHOWN)	PURCHASED
29	1	M22759/16-18-2	INPUT JUMPER WIRE 18 AWG, ~60 MM EACH (SEE NOTE 2) (NOT SHOWN)	PURCHASED
30	1	M22759/16-20-2	INPUT JUMPER WIRE 20 AWG, ~60 MM EACH (SEE NOTE 2) (NOT SHOWN)	PURCHASED
31	19	RS 287-3915	HEX BOLT M4 X 8, INTO THE PEM NUT IN THE STRIP	A2 STAINLESS
32	19	RS 231-0691	SPRING WASHER M4 UNDER EACH JUMPER BOLT HEAD	A2 STAINLESS
33	2	Klauke 2R5	TUBULAR LUG 10 MM2, M5 HOLE	Cu, TINNED
34	2	RS 287-3921	HEX BOLT M5 X 10, FEED LUG INTO THE PEM SS-M5…	A2 STAINLESS
35	4	EFC 10S00056	COMPRESSION LIMITER, FEET (4)	STEEL, ZINC

NOTES:

1. OVERALL SIZES ARE FOR REFERENCE.
2. PARTS THAT SHARE ONE MODEL (THE MICRO RELAYS, LUGS, JUMPER WIRES) ARE BALLOONED ON THE FIRST ITEM OF THEIR GROUP.
3. USE THE M5 x 10 FEED BOLT, NOT A SHORTER ONE: ITS TIP GOES 1.7 INTO THE SPINE AND LOCKS THE STRIP.
4. SHEET 2: WHAT EACH FUSE AND RELAY FEEDS. SHEETS 3 AND 4: THE HARNESS WIRE AT EACH TERMINAL, NAMED AS ON THE HARNESS DRAWINGS.
5. TERMINALS GO IN FROM UNDER THE DECK, CONTACT BOX FIRST, UNTIL THE LANCE CLICKS. PULL THE WIRE BACK GENTLY TO CHECK IT HOLDS (TE 114-2025 3.6, 114-20021 3.7).
6. KEEP HARNESS TIES AT LEAST 38 mm BELOW THE DECK, AND DO NOT BEND THE WIRES HARD TO ONE SIDE AT THE DECK (TE 114-20021 3.7 C).

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
LOGIC FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100L	V	SEE BOM

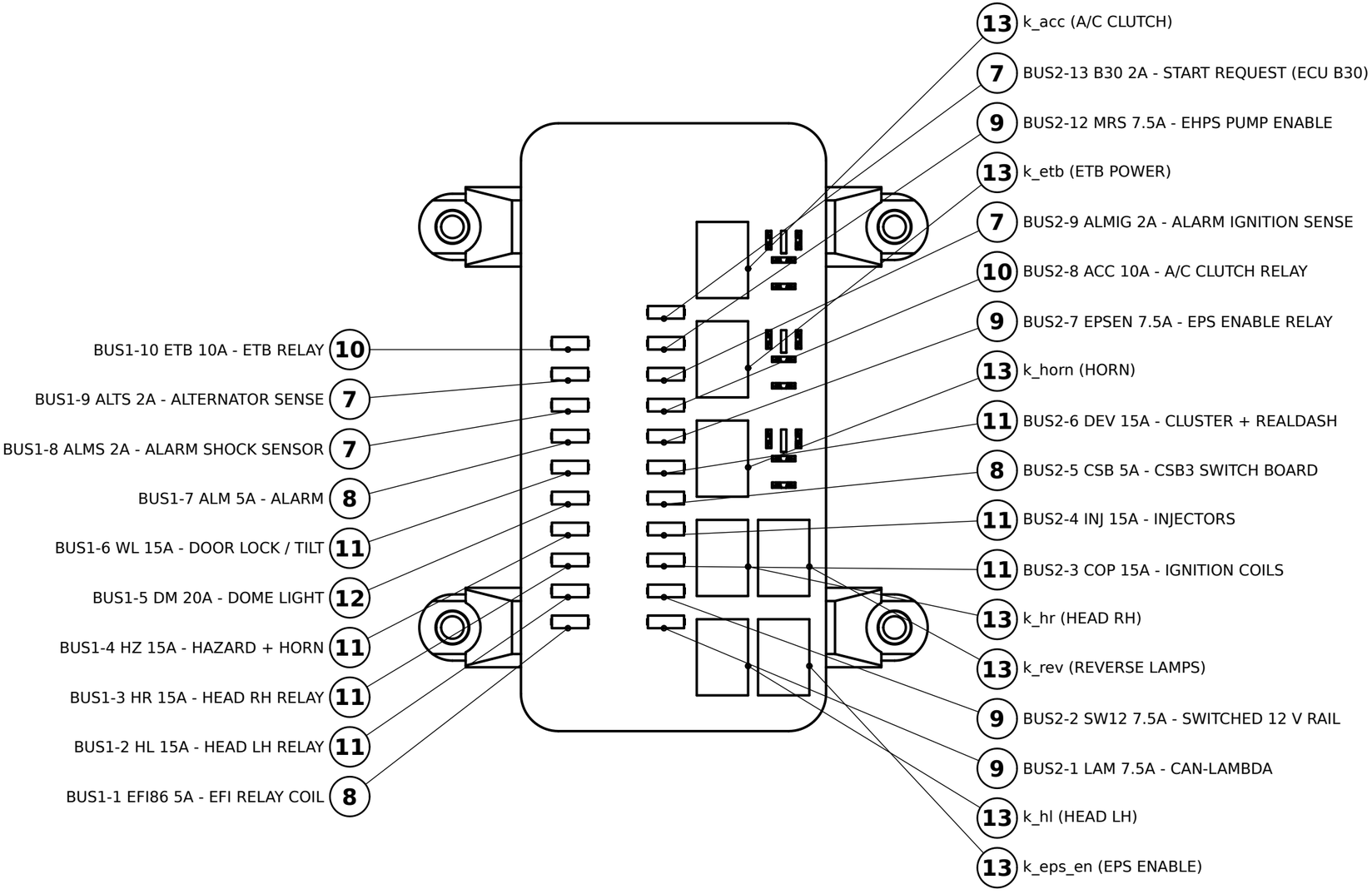
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SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

WHAT EACH FUSE AND RELAY FEEDS (COVER OFF)

TOP VIEW SCALE 1:1.5



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
LOGIC FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100L	V	SEE BOM

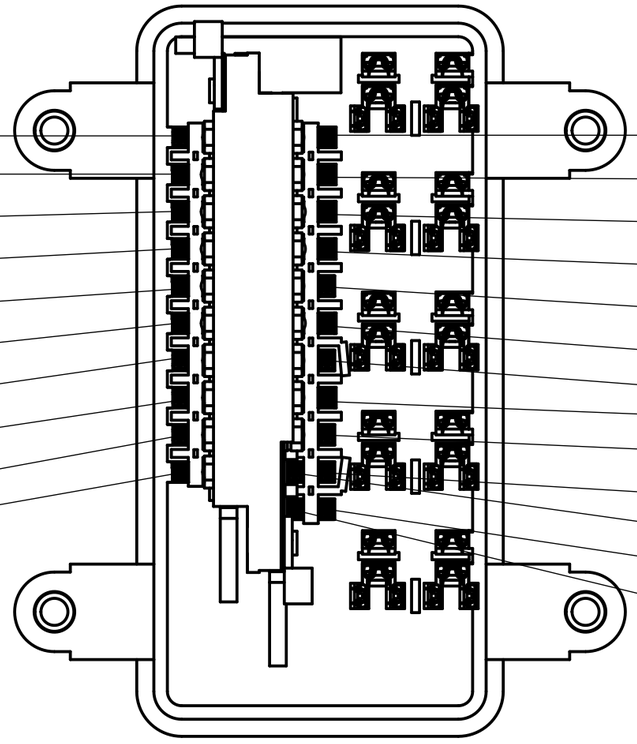
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SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

HARNESS WIRES AT THE FUSES - SEEN FROM BELOW, BELLY PAN OFF

BOTTOM VIEW SCALE 1:1.5



BUS1-1 > POWER R2 86 - EFI coil+

BUS1-2 > HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86

BUS1-3 > HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86

BUS1-4 > HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay

BUS1-5 > OEM J/B 1 plug 1H pin 2 (LH kick), L-Y wire - DOME 20A feed (EWD 1993 p.132)

BUS1-6 > OEM old R/B2 relay lead pin 2 (LH kick), W-L wire - door lock + tilt-away (EWD p.137)

BUS1-7 > Alarm +12 V splice (AntiTheft loom, center dash) - alarm pin 1 constant +12 V

BUS1-8 > Shock sensor +12 V splice (AntiTheft loom, ECU area) - shock sensor pin A

BUS1-9 > Alternator plug G2 pin 1 (engine bay, via loom C) - S battery sense (EWD 1993 p.74-75)

BUS1-10 > LOGIC M14 30 - ETB 12V in

BUS2-1 > Bulkhead B pin 95 (DRB102, RH firewall) - CAN-Lambda +12 V

BUS2-2 > SW12 bus plug DT06-12 (glove box, by LOGIC box) 1 - SW12 +12 V bus

BUS2-3 > Bulkhead A pin 2 (DRB102, RH firewall) - COP coil +12 V rail (COP 15A)

BUS2-4 > Bulkhead A pin 1 (DRB102, RH firewall) - injector +12 V rail (INJ 15A)

BUS2-5 > CSB3 plug pin 1 (CSB3 box under the cluster) - CSB3 +12 V switched

BUS2-6 > Dash device +12 V splice (CAN loom) - center cluster and RealDash Pi +12 V

BUS2-7 > LOGIC M21 30 - EPS EN 12V in

BUS2-8 > LOGIC M15 30 - A/C 12V in

BUS2-9 > Alarm ignition splice (AntiTheft loom, center dash) - alarm pin 3 ignition sense

BUS2-12 > Bulkhead A pin 42 (DRB102, RH firewall) - EHPS pump logic enable (pump pin C1)

BUS2-12 IN > LOGIC M21 87 - EPS EN 12V out

BUS2-13 > ECU plug B pin 30 (Link ECU, center dash) - DI 7 start request

BUS2-13 IN > OEM starter relay socket R/B4 pin 1, B-W wire - ignition ST1 start request (EWD 1993 p.66)

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
LOGIC FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100L	**V**	SEE BOM

SCALE	UNITS	SIZE	SHEET	DATE	DRAWN
1:1.5	mm	B (11x17)	3 OF 6	2026-10-19	SteveCAD

SOURCE
C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

HARNESS WIRES AT THE RELAYS - SEEN FROM BELOW, BELLY PAN OFF

BOTTOM VIEW SCALE 1:1.5

k_hl 30 > HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86

k_hl 87 > LH headlight plug H1 pin 3 (loom C, LH front) - headlight LH +12 V (EWD 1993 p.116)

k_hl 86 > HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86

k_hl 85 > Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2

k_hr 30 > HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86

k_hr 87 > RH headlight plug H2 pin 3 (loom C, RH front) - headlight RH +12 V (EWD 1993 p.116)

k_hr 86 > HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86

k_hr 85 > Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2

k_horn 30 > HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay

k_horn 87 > Horn splice (loom C, front) - horn relay output to horns H3 and H4 pin 1

k_horn 86 > HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay

k_horn 85 > OEM cowl plug IE1 pin 19 (LH kick), G wire - horn switch line (EWD 1993 p.185)

k_etb 30 > FB-LOGIC BUS1-10 - ETB 10A

k_etb 87 > ECU plug B pin 5 (Link ECU, center dash) - ETB motor supply from the ETB relay

k_etb 85 > ECU plug A pin 20 (Link ECU, center dash) - Aux 2 sinks the ETB relay coil

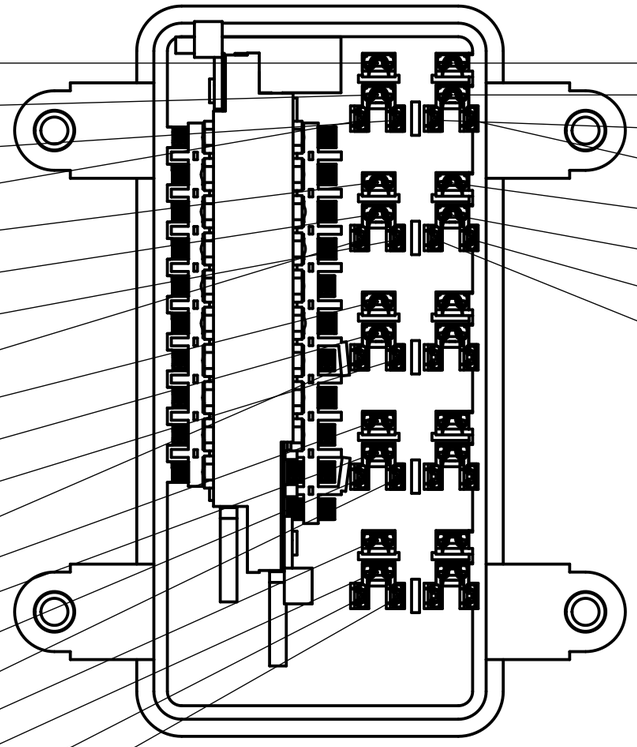
k_etb 86 > SW12 bus plug DT06-12 (glove box, by LOGIC box) 5 - SW12 +12 V bus

k_acc 30 > FB-LOGIC BUS2-8 - ACC 10A

k_acc 87 > A/C magnet clutch A4 pin 4 (loom C, compressor) - clutch +12 V (1990 EWD p.152)

k_acc 85 > ECU plug B pin 10 (Link ECU, center dash) - Ign 8 sinks the A/C clutch relay coil

k_acc 86 > A/C pressure switch splice (A-cabin loom) - switch A5 pin 4 output to the clutch relay



k_eps_en 30 > FB-LOGIC BUS2-7 - EPSEN 7.5A

k_eps_en 87 > FB-LOGIC BUS2-12 IN - MRS in

k_eps_en 85 > ECU plug B pin 12 (Link ECU, center dash) - Ign 6 sinks the EPS enable relay coil

k_eps_en 86 > SW12 bus plug DT06-12 (glove box, by LOGIC box) 9 - SW12 +12 V bus

k_rev 30 > GAUGE 15A splice - J/B 3 feed to reverse relay pin 30 and the back-up switch

k_rev 87 > Back-up lamp splice (RearLights loom, rear) - R11 and R12 pin 4 back-up lamps

k_rev 86 > Reverse switch splice (B-cabin loom) - back-up switch B1 pin 1, +12 V in reverse

k_rev 85 > POWER box NEG bar M8 stud (glove box) - reverse relay coil ground

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
LOGIC FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100L	**V**	SEE BOM

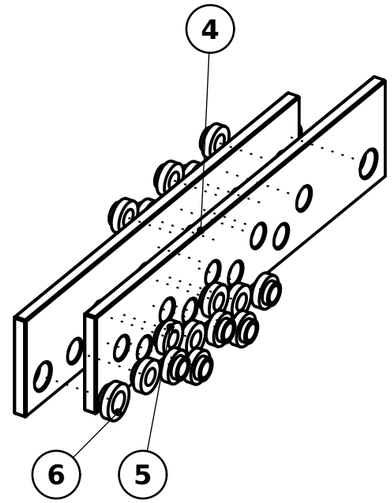
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3RD ANGLE

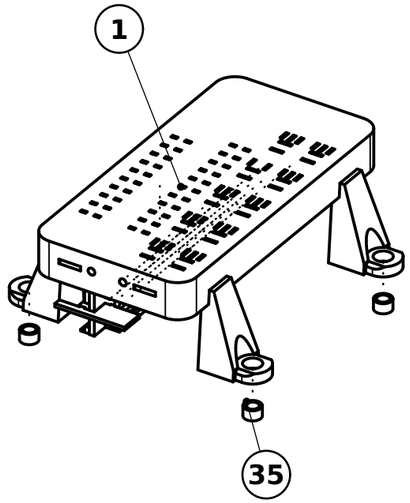
STEP 1: PRESS A PEM NUT INTO EVERY STRIP HOLE

EXPLODED TRIMETRIC (NOT TO SCALE)



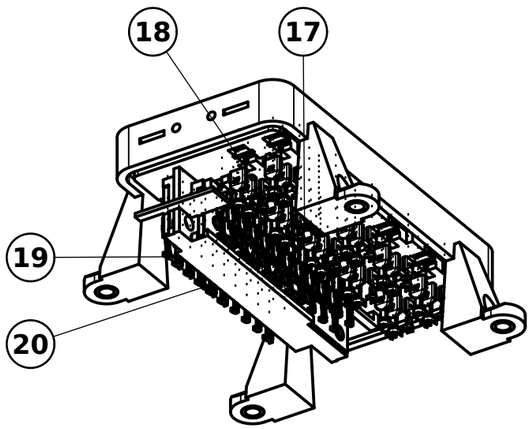
STEP 2: LIMITERS; STRIPS IN FROM THE REAR TILL THE HOOK CLICKS

EXPLODED TRIMETRIC (NOT TO SCALE)



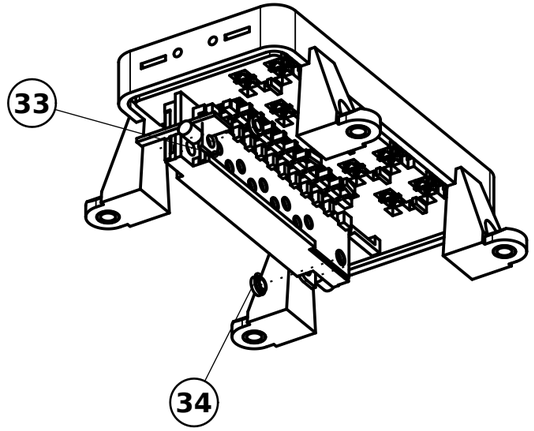
STEP 3: TERMINALS UP INTO THE CAVITIES

EXPLODED, SEEN FROM BELOW (NOT TO SCALE)



STEP 4: FEED LUGS ON THE M5 x 10 FEED BOLTS

EXPLODED, SEEN FROM BELOW (NOT TO SCALE)



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
LOGIC FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100L	**V**	SEE BOM

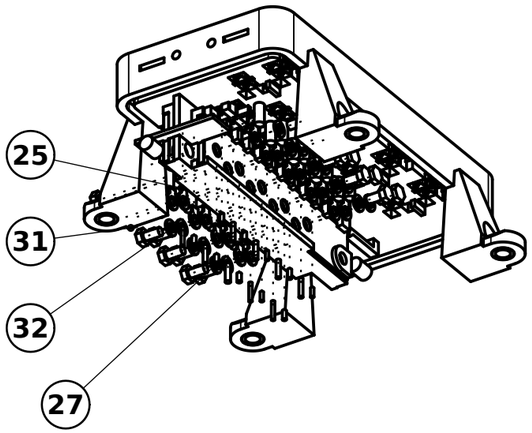
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SOURCE
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3RD ANGLE

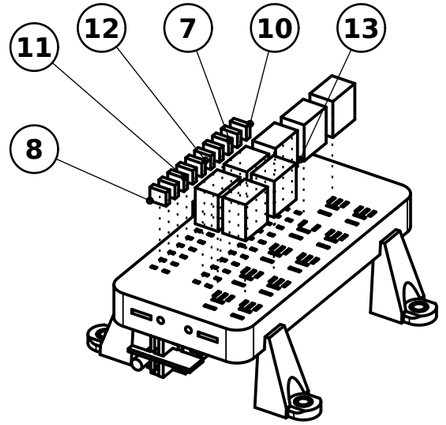
STEP 5: JUMPERS: WIRE, LUG, WASHER, M4 x 8 BOLT

EXPLODED, SEEN FROM BELOW (NOT TO SCALE)



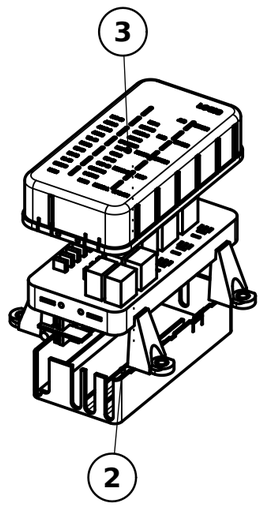
STEP 6: FUSES AND RELAYS IN FROM THE TOP

EXPLODED TRIMETRIC (NOT TO SCALE)



STEP 7: BELLY PAN AND COVER

EXPLODED TRIMETRIC (NOT TO SCALE)



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
LOGIC FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100L	**V**	SEE BOM

SCALE	UNITS	SIZE	SHEET	DATE	DRAWN
AS SHOWN	mm	B (11x17)	6 OF 6	2026-10-19	SteveCAD

SOURCE
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3RD ANGLE